LITHIUM

Narrow therapeutic index

Toxicity with dehydration/NSAIDs

CHEMICAL NAME Dilithium carbonate (Li_2CO_3); also lithium citrate
CLASS Mood stabiliser
TRADE NAMES Priadel · Camcolit · Lithobid · Liskonum

THERAPEUTIC USES / INDICATIONS

- Bipolar disorder: prophylaxis (first-line), acute mania.
- Recurrent depression (augmentation).
- Reduces suicide risk in mood disorders; aggressive/self-harming behaviour.

ADMINISTRATION

- **PO** usually once daily at night (MR); prescribe by brand (bioavailability differs).
- Lithium citrate liquid and carbonate tablets not mg-equivalent.
- Levels taken 12 h post-dose.

DOSAGES

- **Adult:** start 200–400 mg at night, adjust to level.
- **Target level:** 0.4–1.0 mmol/L (0.6–0.8 maintenance; up to 1.0 mania; 0.4–0.6 elderly).
- Levels weekly after start/changes until stable, then every 3 months.

PHARMACOKINETICS

- **Onset:** 1–3 weeks (antimanic).
- Not metabolised; not protein-bound; distributed in total body water.
- Excreted unchanged by kidney (reabsorbed with Na^+ — dehydration/↓ Na^+ ↑ levels). **t½:** 18–36 h.

PHARMACODYNAMICS

- Mechanism not fully known: inhibits **inositol monophosphatase** and **GSK-3β**, modulates neurotransmission and neuroprotection.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Fine tremor, thirst, polyuria (nephrogenic DI).
- Nausea, diarrhoea (early).
- Weight gain, metallic taste, oedema.

ADVERSE EFFECTS (SERIOUS)

- Toxicity (>1.5 mmol/L): coarse tremor, vomiting, diarrhoea, ataxia, slurred speech, confusion, seizures, coma, arrhythmias.
- Hypothyroidism; hyperparathyroidism/hypercalcaemia.
- Chronic kidney disease; nephrogenic diabetes insipidus.
- Teratogenic (Ebstein's anomaly); worsening psoriasis/acne.

CONTRAINDICATIONS & CAUTIONS

- Significant renal impairment; cardiac disease (sick sinus); untreated hypothyroidism.
- Addison's disease; low-sodium diet; dehydration.
- Brugada syndrome.
- Pregnancy (relative; specialist), breastfeeding.

KEY INTERACTIONS

- ↑ lithium levels: NSAIDs, ACE inhibitors, ARBs, thiazide (and loop) diuretics, metronidazole.
- **Theophylline, sodium bicarbonate:** ↓ levels.
- **Antipsychotics (haloperidol):** neurotoxicity; **SSRIs:** serotonin syndrome.

NURSING MANAGEMENT

- Levels 12 h post-dose: weekly initially, then 3-monthly.
- U&E, eGFR, TFTs, calcium, weight every 6 months; ECG baseline.
- Assess for toxicity signs; hold dose and check level if suspected.
- Ensure fluid intake 2–3 L/day; stop during dehydrating illness.
- Lithium alert card and booklet.

PATIENT EDUCATION

- Drink enough fluids and keep salt intake steady; take care in hot weather, vomiting, diarrhoea.
- Avoid NSAIDs (ibuprofen) — use paracetamol.
- Report vomiting, diarrhoea, coarse tremor, unsteadiness, confusion; stop and seek help.
- Plan pregnancy with your psychiatrist; same brand always.

KEY POINTS

- Mood stabiliser — target 0.4–1.0 mmol/L (12 h level).
- NSAIDs, ACEi, diuretics, dehydration ↑ levels.
- Toxicity: coarse tremor, vomiting, ataxia, confusion.
- Thyroid and kidney monitoring.